

University of Central Punjab -FOIT

COAL Fall 2025 – C13

Quiz 3

Total Marks = 3+2+2+3 = 10

CLO = 2

1. A programmer writes the following ISR:

```
myISR:
    inc cx
    mov ax, cx
    iret
```

The interrupt was invoked while the main program was using CX for loop counting.

- a) What problem can this cause in the main program?

CX is not preserved before using inside ISR.

- b) Write the instruction(s) that should be added to avoid this issue.

Push CX

Pop CX

- c) Why is that instruction necessary?

To preserve the state of CX to run the loop in main program

2. How does a computer system ensures that while handling an interrupt, CPU does not handle any other interrupts? How does the system go back to its previous state after interrupt has been handled?

By disabling IF and TF before transferring control to ISR. Both flags are restored once the interrupt has been handled successfully.

3. What is the interrupt number whose IVT entry starts at physical address 0000:009Ch?

Interrupt number = 27h

Reason: $009Ch \div 4 = 27h$.

4. The following code assembles correctly, but does not work as intended:

```
mov ax, 0
mov es, ax
mov word ptr es:[21h*4], offset new21
mov word ptr es:[21h*4+2], cs
new21:
    mov ah, 2
    mov dl, '*'
    int 21h
    iret
```

Identify, explain (one line) and correct the problem in this code.

Problem: Calling INT 21h inside a custom INT 21h handler causes infinite recursion.

Reason: ISR re-invokes the same interrupt it hooks.

Correction: use int 10 to display.